Hostel Management System

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Abstract - Hostel Management System

Project Overview:

A comprehensive desktop application developed using Python (CustomTkinter) and MySQL that performs essential hostel operations such as room management, student registration, room allocation, and billing through a modern interactive user interface.

Purpose:

To automate and streamline common hostel administration tasks, demonstrating practical database integration, CRUD operations, and GUI development capabilities.

Core Features:

- Manage multiple hostel rooms and capacities
- Register students and maintain their records
- Allocate rooms based on availability
- Track occupancy and billing status
- Interactive dashboard for real-time statistics
- Export allocations data to CSV

Technologies Used:

- Python (CustomTkinter for GUI)
- MySQL (Database Backend)
- mysql-connector-python (Database connection)
- CSV Module (Data export)

Problem Description & System Design

Problem Description:

Managing hostel accommodations, student information, and billing manually or through basic spreadsheets can be error-prone and time-consuming. In larger hostels, tracking room availability and ensuring fees are paid on time requires significant effort. There is a clear need for a locally hostable, user-friendly tool that consolidates these operations into a unified interface, allowing administrators to process data efficiently.

System Design / Workflow:

1. Start Application: Administrator opens the GUI application.

2. Database Initialization: Application automatically connects and provisions necessary MySQL tables.

3. User Interaction: The administrator navigates the tool using the sidebar buttons (Add Room, Register, Allocate).

4. Process Request: Input data is serialized and dispatched via SQL queries to the backend.

5. Result Retrieval: Results such as updated seat allocations and dashboards are rendered dynamically in the interface.

System Interface Screenshots

This section presents annotated screenshots of the HostelPro web application developed as part of the Hostel Room Allocation System. The application uses a React.js frontend with a FastAPI (Python) backend and a MySQL database.

Figures included:

- Fig. 1: Dashboard Overview
- Fig. 2: Rooms Inventory Management
- Fig. 3: Student Registration
- Fig. 4: Room Allocations
- Fig. 5: Fee Management

Fig. 1: Dashboard Overview

The Dashboard displays real-time statistics including Total Rooms, Occupancy, Registered Students, and Pending Fees. The Quick Overview panel below shows total capacity, occupied beds, and available slots.

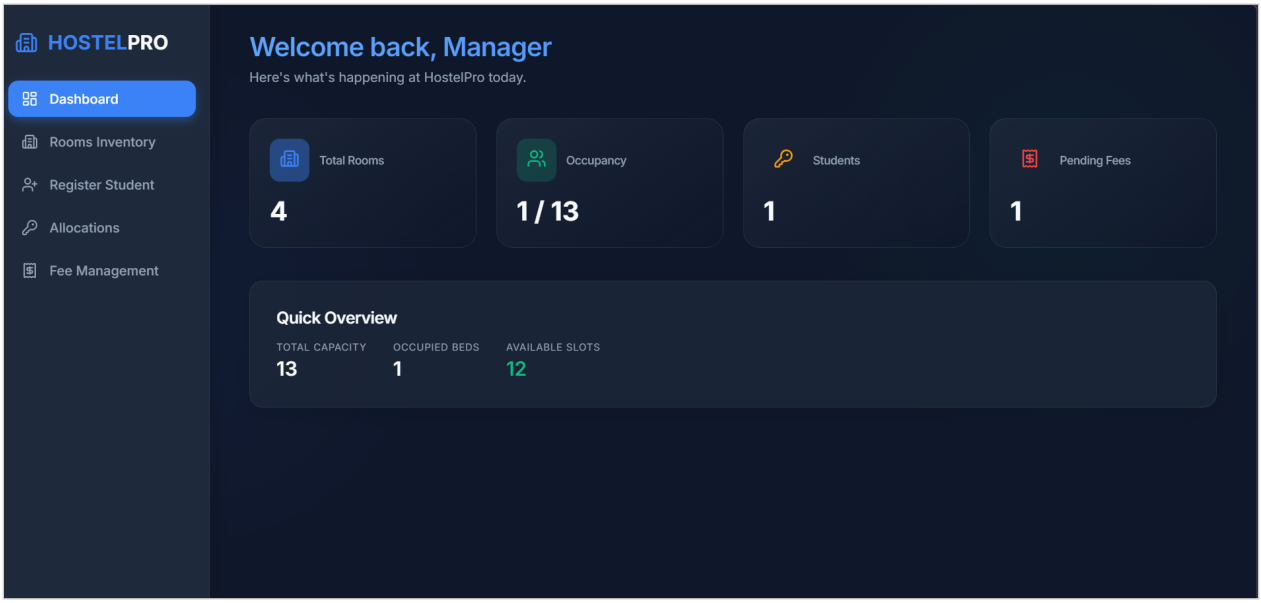


Fig. 2: Rooms Inventory Management

The Rooms Inventory page allows administrators to add rooms by specifying a room number, maximum capacity, and availability status. Existing rooms are listed with live occupancy ratios and a Delete option.

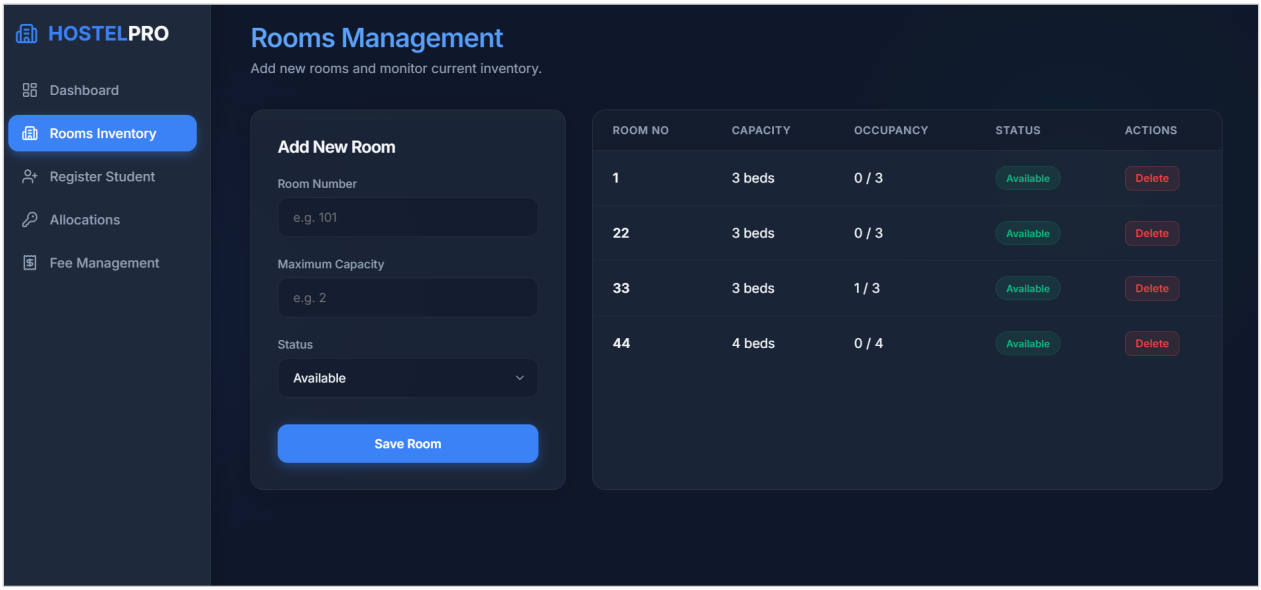


Fig. 3: Student Registration

The Register Student page allows new students to be enrolled by providing their full name and contact details. All registered students are displayed in a list below the form.

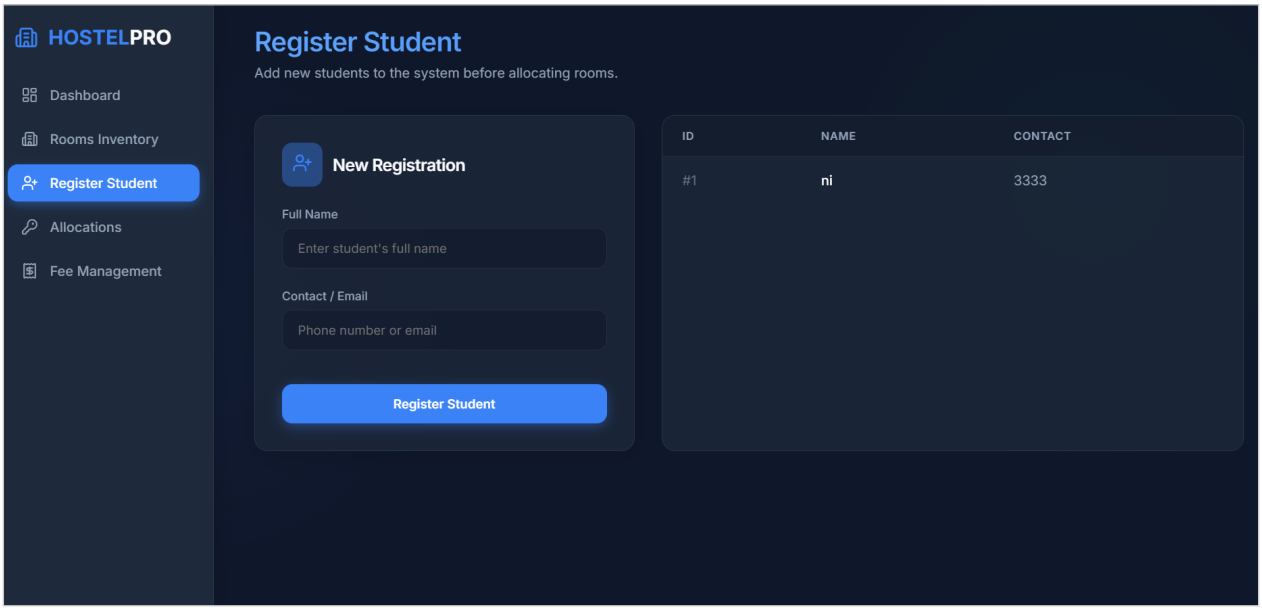


Fig. 4: Room Allocations

The Room Allocations page enables managers to assign an available room to a student along with a monthly rent amount. All current allocations are shown in a table with the date of allocation and a Revoke option.

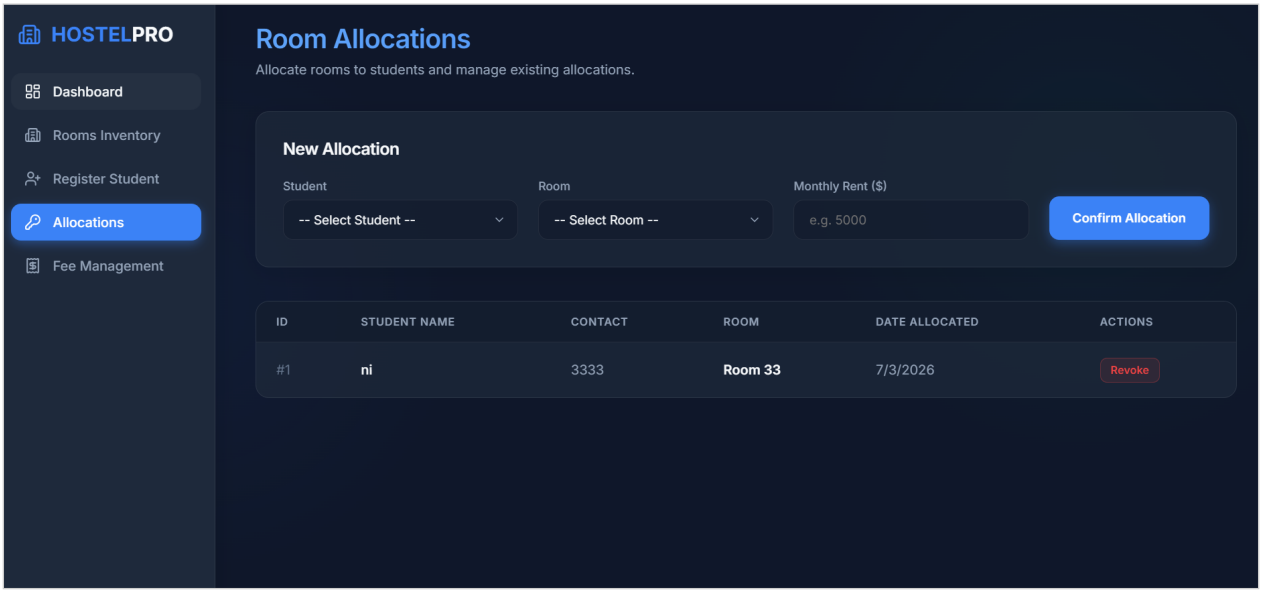
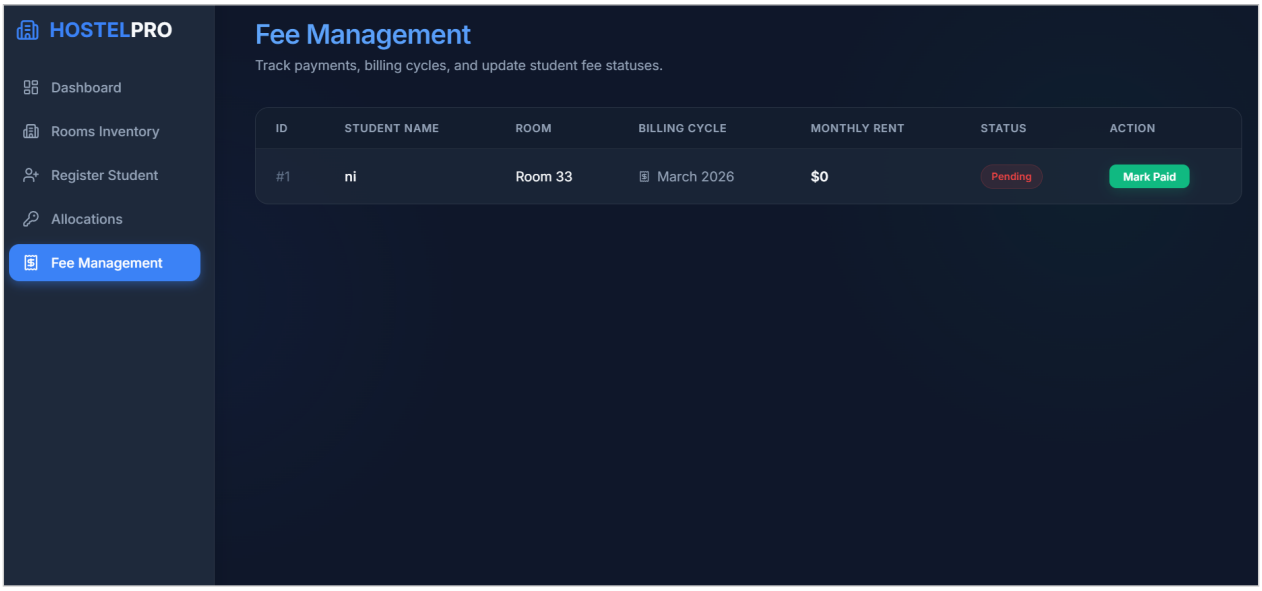


Fig. 5: Fee Management

The Fee Management page tracks each student's billing cycle, monthly rent, and payment status. Administrators can mark individual fees as Paid with a single click.



Conclusion & Scope for Extension

Conclusion:

The project successfully achieved its goal of creating a secure, efficient, and user-friendly Hostel Room Allocation System. The application transitions from a Python desktop (tkinter/CustomTkinter) application to a modern full-stack web application built with React.js on the frontend and FastAPI (Python) on the backend, backed by a MySQL relational database.

The system enables hostel administrators to manage rooms, register students, allocate rooms, and track fee payments — all through a streamlined web interface with real-time statistics on the dashboard.

Scope for Extension:

The following enhancements can be considered for future development:

• Student self-registration portal with login authentication.
• Email/SMS notifications for fee due dates and allocation confirmations.
• Export reports as PDF or Excel for administrative record-keeping.
• Room preference selection and waitlist management.
• Developing a cloud-ready deployment version using Docker and CI/CD pipelines.
• Integration with a payment gateway for online fee collection.

References

- FastAPI Documentation: https://fastapi.tiangolo.com/
- React.js Documentation: https://react.dev/
- MySQL Documentation: https://dev.mysql.com/doc/
- ReportLab Documentation: https://docs.reportlab.com/
- pypdf Documentation: https://pypdf.readthedocs.io/
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